

Cocomo Model :

Constructive cost model (Cocomo Model) used for calculate the effort, development time, Average staff size and productive.

Three Types of Cocomo Models :

- I. Basic Cocomo Model ,
- II. Intermediate Cocomo Model,
- III. Complete\detailed Cocomo Model.

Basic Cocomo Model :

- ✓ It estimates the software in a rough and quick manner .
- ✓ Mostly useful for small-medium sized software .
- ✓ 3 Modes Of development
 - I. Organic
 - II. Semidetached
 - III. Embedded

	Organic	Detached	Embedded
1.Size	2-50 kLOC	50-300 KLOC	Over 300 KLOC
2.Team Size	Small size	Medium size	Large size
3.Developer Experience	Experienced developers needed	Average Experience people	Very little previous experience
4. Environment	Familiar environment	Less familiar	Significant changes (almost new)
5.Inovation	Little	Medium	Major
6.Deadline	Not tight	Medium	Tight

Basic model equation :

1. Effort = a (KLOC)^b person-months
2. Development = c (Effort)^d months
3. Average Staff = Effort /Dev:Time person
4. Productivity = KLOC/Effort KLOC/P.M

Mode	a	b	c	d
Organic	2.4	1.05	2.5	0.38
Semi-detached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.32

Q : Suppose that a project was estimate to be 400 KLOC . Calculate effort and Time for each of 3 modes of development .

1. **Organic** :

$$\begin{aligned}\text{Effort} &= a (\text{KLOC})^b \quad \text{person-months} \\ &= 2.4 (400)^{1.05} = 1295 \text{ P.M}\end{aligned}$$

$$\begin{aligned}\text{Development time} &= c (\text{Effort})^d \text{ months} \\ &= 2.5 (1295) ^{0.38} \text{ M} \\ &= 38 \text{ M}\end{aligned}$$

$$\begin{aligned}\text{Average staff} &= \text{Effort}/\text{Dev:time} \text{ person} \\ &= 1295/38 = 34.07 \text{ person}\end{aligned}$$

$$\begin{aligned}\text{Productivity} &= \text{KLOC}/\text{Effort} \quad \text{KLOC/P.M} \\ &= 400/1295 \text{ KLOC/P.M} \\ &= 0.3089 \text{ kloc/p.m}\end{aligned}$$

2. **Semi-detached**

$$\begin{aligned}\text{Effort} &= a (\text{KLOC})^b \quad \text{person-months} \\ &= 3.0 (400)^{1.12} = 2462.79 \text{ P.M}\end{aligned}$$

$$\begin{aligned}\text{Development time} &= c (\text{Effort})^d \text{ months} \\ &= 2.5 (2462.79) ^{0.35} \text{ M} \\ &= 38.45 \text{ M}\end{aligned}$$

$$\begin{aligned}\text{Average Staff} &= \text{Effort} / \text{Dev:Time} \text{ persons} \\ &= 2462.79/38.45 \text{ persons} \\ &= 64.05 \text{ persons}\end{aligned}$$

$$\begin{aligned}\text{Productivity} &= \text{KLOC}/\text{Effort} \quad \text{KLOC/P.M} \\ &= 400/2462.79 \text{ KLOC/P.M} \\ &= 0.1624 \text{ kloc/p.m}\end{aligned}$$

1. Embedded :

$$\begin{aligned}\text{Effort} &= a (\text{KLOC})^b \quad \text{person-months} \\ &= 3.6 (400)^{1.20} = 4772 \text{ P.M}\end{aligned}$$

$$\begin{aligned}\text{Development time} &= c(\text{Effort})^d \text{ months} \\ &= 2.5(4772)^{0.32} \text{ M} \\ &= 37.5\text{M}\end{aligned}$$

$$\begin{aligned}\text{Average Staff} &= \text{Effort} / \text{Dev:Time} \text{ persons} \\ &= 4772 / 37.5 \text{ persons} \\ &= 127.25 \text{ persons}\end{aligned}$$

$$\begin{aligned}\text{Productivity} &= \text{KLOC} / \text{Effort} \quad \text{KLOC/P.M} \\ &= 400 / 4772 \text{ KLOC/P.M} \\ &= 0.0839 \text{ kloc/p.m}\end{aligned}$$

2. Intermediate Model:

- ✓ Extension of basic
- ✓ Set of 15 additional predictors (cost drivers)

Cost drivers adjust nominal cost of project to actual project environment .

15 cost drivers is divided into a four types :

1. Product Attributes :

- a) Required software Reliability (RELY)
- b) Database Size (DATA)
- c) Product Complexity (CPLX)

2.Computer Attributes

- a) Execution Time Constraints (TIME)
- b) Main Storge Constraints (STOR)
- c) Virtual Machine Volatility (VIRT)
- d) Computer Turnaround Time (TURN)

3.Personal Attributes :

- a) Analyst Capability (ACAP)
- b) Application Experience (AEXP)
- c) Programmer Capability (PCAP)
- d) Programming Language Experience (LEXP)

4.Project Attributes :

- a) Moder Programming Practices (MODP)
- b) Use of software Tools (TOOL)
- c) Required Development Schedude (SCED)

Each Cost drivers is rated for the given project environment :

- I. Very low
- II. Low
- III. Nominal
- IV. High
- V. Very high

VI. Extra high

Cost Drivers	Very Low	Low	Nominal	High	Very High
Product Attributes					
Required Software Reliability	0.75	0.88	1.00	1.15	1.40
Size of Application Database		0.94	1.00	1.08	1.16
Complexity of The Product	0.70	0.85	1.00	1.15	1.30
Hardware Attributes					
Runtime Performance Constraints			1.00	1.11	1.30
Memory Constraints			1.00	1.06	1.21
Volatility of the virtual machine environment		0.87	1.00	1.15	1.30
Required turnabout time		0.94	1.00	1.07	1.15
Personnel attributes					
Analyst capability	1.46	1.19	1.00	0.86	0.71
Applications experience	1.29	1.13	1.00	0.91	0.82
Software engineer capability	1.42	1.17	1.00	0.86	0.70
Virtual machine experience	1.21	1.10	1.00	0.90	
Programming language experience	1.14	1.07	1.00	0.95	
Project Attributes					
Application of software engineering methods	1.24	1.10	1.00	0.91	0.82
Use of software tools	1.24	1.10	1.00	0.91	0.83
Required development schedule	1.23	1.08	1.00	1.04	1.10

Effort Adjustment Factor (EAF) :

Calculated by multiplying all the values that have been obtained after categorizing each cost drivers .

Equation for Cocomo :

$$\text{Effort} = a_i(\text{KLOC})^{b_i} * \text{EAF}$$

$$\text{Development Time} = c_i (\text{Effort})^{d_i}$$

Project	ai	bi	ci	di
Organic	3.2	1.05	2.5	0.38
Semidetached	3.0	1.12	2.5	0.35
Enbedded	2.8	1.20	2.5	0.32

Q: A project size of 700 KLOC is to be developed .The Software development team has average experience on similar types of projects .

The project schedule is medium and the effort adjustment factor is 36.5. Calculate the effort ,development time , and person requirements for each of the three modes i.e. organic, semidetached and embedded for the following coefficient :

Organic :

$$\text{Effort} = a_i(\text{KLOC})^{b_i} * \text{EAF}$$

$$= 3.2(700)^{1.05} * 36.5 \quad \text{p.m}$$

$$= 113447.7 \text{ p.m}$$

$$\text{Development Time} = c_i (\text{Effort})^{d_i}$$

$$= 2.5 (113447.7)^{0.38} = 638.67 \text{ M}$$

Semidetached :

$$\begin{aligned}\text{Effort} &= a_i(\text{KLOC})^{b_i} * \text{EAF} \\ &= 3.0(700)^{1.12} * 36.5 \text{ p.m} \\ &= 168237.933 \text{ p.m}\end{aligned}$$

$$\begin{aligned}\text{Development Time} &= c_i (\text{Effort})^{d_i} \\ &= 2.5(168237.933)^{0.35} = 168.67 \text{ M}\end{aligned}$$

Embedded :

$$\begin{aligned}\text{Effort} &= a_i(\text{KLOC})^{b_i} * \text{EAF} \\ &= 2.8(700)^{1.20} * 36.5 \text{ p.m} \\ &= 265196.96 \text{ p.m}\end{aligned}$$

$$\begin{aligned}\text{Development Time} &= c_i (\text{Effort})^{d_i} \\ &= 2.5(265196.96)^{0.32} = 135.98 \text{ M}\end{aligned}$$